Remember to format the output with decimals aligned in columns.

1. The input to the problem is the quantity of widgets. Your program should determine the price to charge based on the schedule below. Calculate the extended price (quantity x price). Calculate tax at 7%. Display the extended price, tax amount and total.

Quantity	Price
>10,000	$10
5,000 to 10,000	$20
Below 5,000	$30

input	process	output
Int quantity Real price Real tax Real total Real extprice	If quantity < 5,000 Price = 30 Else If quantity < 10,000 Price = 20 else Price = 10 Extprice = price * quantity Tax = extprice*.07 Total = tax+extprice	‘Extended Price ‘ extprice ‘Tax ‘ tax ‘Total ‘ total Spaced accordingly

2. Enter a part number and quantity Determine the cost per unit using the table below. Then calculate the total cost (quantity x unit cost). Display the part number, cost per unit and total cost. Note: Part number can be an integer but it can also be a string because you are not doing arithmetic on it. However, in your code if statement be sure to compare using consistency, that is, if item == “10” when item is a string and if item == 10 when item is an integer.

Part	Unit Cost
10 or 55	1.00
99	2.00
80 or 70	3.00
All others	5.00

input	process	output
Int partnum Int quantity Real total Real unitp	If partnum == 10 or 55 then, Unitp = 1.00 Else If partnum == 99, then Unitp = 2.00 Else If partnum == 80 or 70 then, Unitp = 3.00 Else	‘Part number’ partnum ‘Quantity ‘ quantity ‘Total ‘ total ‘Unit price’ unitp

	Unitp = 5.00 Total = quantity*unitp	

3. Enter a principle amount of a CD and year to maturity of CD. Determine the interest rate based on the amount of the principle and maturity (see below). Calculate first year interest (principle x interest rate). Display principle, interest rate and the interest amount for the first year.

Principle	Years to Maturity	Interest Rate
>$100,000	5	6%
$50,000 to $100,000	10	5%
$50,000 to $100,000	5	4%
Any other principle	any other years	2%

input	process	output
Real principal Int year Real interest Real fyi (first year interest)	If year == 5 If principal > 100,000 Interest = .06 Else If principal >= 50,000 Interest = .04 else Interest = .02 Else If year == 10 If principal >= 50,000 If principal <= 100,000 Interest = .05 else Interest = .02 else Interest = .02 else Interest = .02 Fyi = principal*interest	‘Principal ‘ principal ‘Interest rate ‘ interest ‘Interest amount ‘ FYI

4. Have the user to enter number of concert tickets. The price per ticket depends on the volume (see below). Display the number of tickets, price per ticket and the total cost (number of tickets x Price Per Ticket).

Quantity	Price Per Ticket
>=25	$50
10 to 24	$60
5 to 9	$70
Less 5	$75

input	process	output
Int tickets Float cost Float price	If tickets <5: Price = 75 else If tickets <=9 Price = 70 else If tickets <=24 Price = 60 else Price = 50 Cost = price*tickets	‘Tickets ‘ tickets ‘Price ‘ price ‘Total cost’ cost

5. The user will enter the employee last name, salary and job level (as noted below). Use the job level to determine the bonus rate. Then compute bonus to be salary times bonus rate. Display employee last name and bonus.

Job Level	Bonus Rate
10 and above	25%
5 to 9	20%
All others	10%

input	process	output
String name Real salary Int level Real bonusRate Real bonus	If level < 5 BonusRate = .10 else If level <10 bonusRate = .20 else BonusRate = .25 Bonus = salary*bonusRate	‘Name ‘ name ‘Bonus amount $‘ bonus